



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited by NAAC with "A+" Grade

SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA

Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

22MA201 – ENGINEERING MATHEMATICS II

UNIT 1 - LP 4 – COOLING / HEATING OF AN OBJECT

1. NEWTON'S LAW OF COOLING / WARMING

Temperature difference in any situation results from heat energy flow into a system or heat energy flow from a system to surroundings. The former leads to heating whereas latter leads to cooling of an object.

The rate at which the temperature of a body changes is proportional to the difference between the temperature of the body and the temperature of the surrounding medium.

Mathematically it is defined as

$$\frac{dT}{dt} \propto (T - T_0)$$

$$\frac{dT}{dt} = -k(T - T_0)$$

where T is the temperature of the body, T_0 is the surrounding temperature (or ambient

temperature) and k is the proportionality constant.

The solution of the above differential equation can be found any one of these following method.

Variable separable method

$$\frac{dT}{dt} \propto (T - T_0)$$

$$\frac{dT}{dt} = -k(T - T_0)$$

$$\frac{dT}{T - T_0} = -k dt$$

$$\int \frac{dT}{T - T_0} = -k \int dt$$

$$\ln(T - T_0) = -kt + \ln c$$

$$T - T_0 = ce^{-kt}$$

$$T = T_0 + ce^{-kt}$$

1.1. Problems

1. A metal bar at a temperature of $100^\circ F$ is placed in a room at a constant temperature of $0^\circ F$. If after 20 minutes the temperature of the bar is $50^\circ F$, then find
 - (i) the temperature of the bar after 10 minutes
 - (ii) the time it will take the bar to reach the temperature of $25^\circ F$

Solution:

Here

$$T_0 = 0$$

$$\text{If } t = 0, T = 100$$

$$\text{If } t = 20, T = 50$$

$$\text{If } t = 10, T = ?$$

$$\text{If } T = 25, t = ?$$

$$\frac{dT}{dt} \propto (T - T_0)$$

$$\frac{dT}{dt} = -k(T - T_0)$$

$$\frac{dT}{dt} = -k(T - 0)$$

$$\frac{dT}{dt} = -kT$$

$$\frac{dT}{T} = -kdt$$

$$\int \frac{dT}{T} = -k \int dt$$

$$\ln T = -kt + \ln c$$

$$T = ce^{-kt} \quad \text{----- (1)}$$

$$\text{At } t = 0, T = 100$$

$$(1) \Rightarrow 100 = ce^0 \Rightarrow c = 100$$

$$(1) \Rightarrow T = 100e^{-kt} \quad \text{----- (2)}$$

$$\text{At } t = 20, T = 50$$

$$(2) \Rightarrow 50 = 100e^{-20k}$$

$$\Rightarrow \frac{1}{2} = e^{-20k}$$

$$\Rightarrow e^{-20k} = \frac{1}{2}$$

$$\Rightarrow -20k = \ln\left(\frac{1}{2}\right)$$

$$\Rightarrow -20k = -0.6931$$

$$\Rightarrow k = 0.0347$$

$$(2) \Rightarrow T = 100e^{-0.0347t} \quad \text{----- (3)}$$

(i) If $t = 10$,

$$T = 100e^{-0.0347(10)}$$

$$T = 70.68^\circ F$$

(ii) If $T = 25$,

$$25 = 100$$

$$0.25 = e^{-0.0347t}$$

$$-0.0347t = -1.3863$$

$$t = 39.95 \text{ min}$$

2. When a cake is removed from an oven, its temperature is measured at $300^\circ F$. Three minutes later its temperature is $200^\circ F$. How long will it take for the cake to cool off to $80^\circ F$ with a room temperature of $70^\circ F$?

Solution:

Here

$$T_0 = 70$$

$$\text{If } t = 0, T = 300$$

$$\text{If } t = 3, T = 200$$

$$\text{If } T = 70, t = ?$$

$$\frac{dT}{dt} \propto (T - T_0)$$

$$\frac{dT}{dt} = -k(T - T_0)$$

$$\frac{dT}{dt} = -k(T - 70)$$

$$\frac{dT}{dt} + kT = 70k$$

$$\text{Here } P = k, Q = 70k$$

$$\text{Integrating factor (I.F.)} = e^{\int P dt} = e^{\int k dt} = e^{kt}$$

The solution is

$$Te^{\int P dt} = \int Qe^{\int P dt} dt + c$$

$$Te^{kt} = \int 70ke^{kt} dt + c$$

$$Te^{kt} = 70k \frac{e^{kt}}{k} + c$$

$$Te^{kt} = 70e^{kt} + c$$

$$T = 70 + ce^{-kt} \quad \text{----- (1)}$$

$$\text{At } t = 0, T = 300$$

$$(1) \Rightarrow 300 = 70 + ce^0 \Rightarrow c = 230$$

$$(1) \Rightarrow T = 70 + 230e^{-kt} \quad \text{----- (2)}$$

$$\text{At } t = 3, T = 200$$

$$(2) \Rightarrow 200 = 70 + 230e^{-3k}$$

$$\Rightarrow \frac{130}{230} = e^{-3k}$$

$$\Rightarrow e^{-3k} = \frac{13}{23}$$

$$\begin{aligned}\Rightarrow -3k &= \ln\left(\frac{13}{23}\right) \\ \Rightarrow -3k &= -0.5705 \\ \Rightarrow k &= 0.1902 \\ (2) \Rightarrow T &= 70 + 230e^{-0.1902t} \quad \text{----- (3)}\end{aligned}$$

If $T = 80$,

$$\begin{aligned}80 &= 70 + 230e^{-0.1902t} \\ 10 &= 230e^{-0.1902t} \\ e^{-0.1902t} &= \frac{1}{23} \\ -0.1902t &= \ln\left(\frac{1}{23}\right) \\ -0.1902t &= -3.1355 \\ t &= 16.49 \text{ min}\end{aligned}$$

3. A heated metal ball is placed in cooler surroundings. Its rate of cooling is 2°C per minute when its temperature is 60°C and 1.2°C per minute when its temperature is 52°C . Find the temperature of the surroundings and the rate of cooling when the temperature of the ball is 48°C .

Solution:

By Newton's law of cooling,

$$\begin{aligned}\frac{dT}{dt} &\propto (T - T_0) \\ \frac{dT}{dt} &= -k(T - T_0) \quad \text{----- (1)}\end{aligned}$$

If $T = 60, \frac{dT}{dt} = 2$, then equation (1) becomes

$$2 = -k(60 - T_0) \quad \text{----- (2)}$$

If $T = 52, \frac{dT}{dt} = 1.2$, then equation (1) becomes

$$1.2 = -k(52 - T_0) \quad \text{----- (3)}$$

From (2) and (3), we get

$$\frac{2}{1.2} = \frac{-k(60 - T_0)}{-k(52 - T_0)}$$

$$2(52 - T_0) = 1.2(60 - T_0)$$

$$52 - T_0 = 0.6(60 - T_0)$$

$$52 - T_0 = 36 - 0.6T_0$$

$$0.4T_0 = 16$$

$$T_0 = 40$$

Sub $T_0 = 40$ in equation (2), we get

$$2 = -k(60 - 40)$$

$$2 = -20k$$

$$k = -0.1$$

If $T = 48$, then $\frac{dT}{dt} = -k(T - T_0)$

$$\frac{dT}{dt} = 0.1(48 - 40)$$

$$\frac{dT}{dt} = 0.8^\circ C \text{ per min}$$

1.1.1. Practice Problems

1. A body at an unknown temperature is placed in a room which is held at a constant temperature of $30^\circ F$. If after 10 minutes the temperature of the body is $0^\circ F$ and after 20 minutes the temperature of the body is $15^\circ F$, find the unknown initial temperature.
2. At 10:07 PM, you find a secret agent murdered. Next to him is a martini that got shaken before he could stir it. Room temperature is $70^\circ F$. The martini warms from $60^\circ F$ to $61^\circ F$ in the 2 minutes from 10:07 pm to 10:09 pm. If the secret agent's martinis are always served at $40^\circ F$, what was the time of death (rounded to the nearest minute)?
3. Suppose that a cup of soup cooled from $90^\circ C$ to $60^\circ C$ after 10 minutes in a room whose temperature was $20^\circ C$.
 - (i) How much longer would it take the soup to cool to $35^\circ C$?
 - (ii) Instead of being left to stand in the room, the cup of $90^\circ C$ soup is placed in a freezer whose temperature is $-15^\circ C$, and it took 5 minutes to cool to $60^\circ C$. How long will it take the soup to cool from $90^\circ C$ to 35° ?

